

Quantifying Metal-Organic Framework (MOF) CO₂ Adsorption Thermal Dynamics

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Introduction

Efficient carbon capture and storage has been a sought after solution within the present and for the oncoming years due to climate change and rising global temperatures. The continuous rise and increase in greenhouse gas emissions, particularly CO₂, has triggered a series of environmental problems, including extreme weather events, forest disturbances, and water erosion [1]. Natural disasters also cause heightened incidents of infectious diseases and intensified rainfall that lead to more property damage and societal blockades. Ever since the 19th century, the abrupt and continual increase of increased from 280 ppm in the early 1800s to 417 ppm in 2022, thus representing an increase of about 50% increase, and by 2100, the intergovernmental Panel on Climate Change claims that the levels can rise up to 980 ppm atmospheric CO₂ [2].

Metal organic frameworks (MOFs) are an upcoming class of materials; trendy and new in the industrial scene, it is known for its tunable, porous lattice-like structure made out of organic linkers, and are known to be useful for energy storage, drug delivery, and, most importantly, carbon capture. Because of the versatility of the MOFs, what to put in the linkers, the different reaction groups, and what metal to put in the capture sites, MOFs have many permutations that can be customizable to the use case. The synthesis and feasibility of producing MOFs can vary between types, but some are faced with high costs, long reaction times, and the need for specialized equipment to achieve high-quality, reproducible structures. Thus the adequate testing of these structures could be expensive and time consuming. With the onset of chemical docking and modeling situations, this cost and experimental need can be mitigated, down-selecting key candidates and research for key circumstances that would be more applicable to the real world than just trying options blindly.

Background

Researchers have been able to model carbon capture MOF interactions through Density Functional Theory (DFT), which provides atomic-level approximations through electronic cloud densities, which can provide accurate insights for molecule interactions that can occur especially at the open metal sites and functionalized linkers. Early tests included both DFT and experimental tests, with ab initio work by Grajciar et al on CuBTC, whereby they paired periodic DFT calculation with microcalorimetry validation [3]. For Mg-MOF-74, one of the most extensively studied MOFs for carbon capture and the primary structure we will be using in our project, Valenzano et al. [4] performed the first periodic DFT-D calculations using the CRYSTAL code combined with variable-temperature infrared spectroscopy, finding a CO₂ adsorption enthalpy of -47 kJ/mol and demonstrating that dispersion forces account for approximately half of the total binding energy at the open Mg²⁺ sites. Subsequent work by Poloni et al. [5] extended this work across a series of Mg-MOF-74 using van der Waals (vdW)-corrected DFT, demonstrating that binding energies are governed by a combination of electrostatics and metal d-orbital interactions, and that vdW-DFT systematically outperforms semi-empirical DFT-D2 for OMS-containing systems. vdW-DFT models are better than conventional DFT models, because vdW-DFT models long range dispersion interactions which are important to modeling non-covalent interactions (especially for OMS sites) while conventional DFT models lack this capability [6].

Despite its accuracy, the computational cost of periodic DFT restricts its application to static, single-configuration binding energy calculations at 0K, lacking the dynamical sampling needed to capture CO₂ adsorption trajectories and thermal effects at realistic to extreme conditions. To bridge this gap, the Open DAC 2023 [7], the dataset compiled over 38 million PBE-D3 (GGA functional with dispersion correction) DFT calculations for CO₂ and H₂O across 8,400 MOFs, providing training data necessary for machine-learned interatomic potentials (MLIPs) capable of operating at DFT accuracy [8]. Building on this, the Universal Models for Atoms (UMA), trained on over 500 million DFT calculations spanning the ODAC, OC20, and OMat24 datasets, providing a DFT surrogate that allows for energy-conserving molecular dynamics at a fraction of the cost of ab initio simulation [9].

In this project, we employ the UMA-ODAC calculator to compute CO₂ adsorption energetics and trajectories within the Mg-MOF-74, combining the chemical accuracy of DFT-trained potentials with the dynamics unavailable to static DFT approaches, and to reduce compute power in order to gain an understanding and simulation of incorporating temperature in molecular dynamics (MD) for MOF systems.

Methods

The model we used is UMA-ODAC MLIP, with ODAC representing “Open Direct Air Capture” dataset/domain and refers to UMA models trained or specialized on adsorption/MOF/direct-air-capture related datasets. The model is designed to predict energies, forces, and absorption interactions over MOFs and its molecular trajectories without retraining, making it ideal for this project [10].

First, we calculated the static binding energies of CO₂ to metal Mg sites of the Mg-MOF-74 MOF. Prior to the adsorption studies, both Mg-MOF-74 framework and free CO₂ molecules were geometry optimized using the BFGS optimizer until forces converged below 0.01 eV/Å.

Binding energies were calculated using:

$$E_{bind} = E_{total} - E_{MOF} - E_{CO_2}$$

Where E_{total} is the relaxed energy of the adsorbed system, E_{MOF} is the relaxed Mg-MOF-74 energy, and E_{CO_2} is the relaxed isolated CO₂ energy. CO₂ molecules were then positioned near each open Mg metal site using multiple approach directions and Mg-O starting distances ranging from 2.2-2.8 Å while avoiding atomic overlap. For each valid placement, the combined MOF-CO₂ system was geometry optimized using BFGS with a convergence threshold of 0.05 eV/Å. Final Mg-O adsorption distances and binding energies were recorded for all Mg sites and visualized through comparative plots.

For the MD portion, we used the ASE coupled with the UMA-ODAC calculator. Canonical ensemble (NVT) molecular dynamics simulations were then conducted using a Langevin thermostat across multiple temperatures to investigate the thermal stability and adsorption behavior of CO₂ within Mg-MOF-74. The simulations used femtosecond-scale timesteps to track total energy, atomic trajectories, and Mg-O adsorption distances. Periodic boundary conditions were implemented to preserve the crystalline framework environment.

MDs were conducted from 300-1000K with $\Delta T = 50K$ steps, trajectory data from the MD simulations were post-processed and we collected binding energy, trajectory heatmaps, and Mg-O adsorption

distances and binding/dwelling time of the CO₂ at each Mg site. At first, we only surveyed up to 0.01 ns, but then realized that the binding energy data is still increasing and or not reaching steady state. Thus, the overall time period was increased to 0.05 ns, where indeed, we do see a steady state behavior that will be discussed more in the next section.

Results and Discussion

Static Binding Energies

The three lowest binding energy sites were around -0.57 eV (-53 kJ/mol), reflecting close to DFT calculations done in previous DFT papers that reported (~-45 kJ/mol), but also falls within the range of -0.5-0.6 eV for DFT models that included van-der-wall corrections, and the average Mg-O binding distance of ~2.35 Å is a bit on the longer end than DFT computed values (~2.2-2.3 Å) [10]. The subtle 8 kJ/mol difference can be attributed to the inherent differences in computation of energetics between DFT and the MLIP, of how they treat relaxation pathways or dispersion treatments differently.

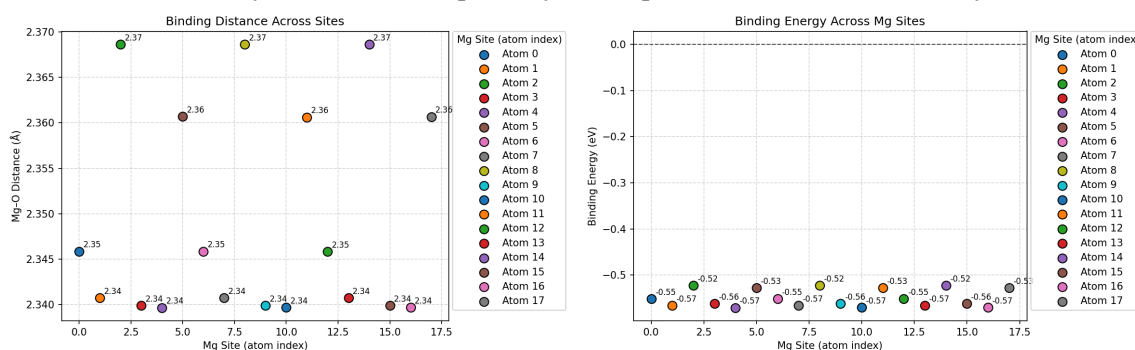


Figure 1: Shown are the static binding Mg-O distance (left) and the static binding energy across all 18 Mg sites.

Molecular Dynamics Simulations

The binding energies we tracked are shown below in Figure 2. There is a black line indicating the neutral binding energy (0 eV) and the red line indicating static binding energy for comparison. Not all of the temperatures were plotted, but four main temperature steps were plotted for clarity. At low temperatures ($T \leq 400$ -450K), the E_{bind} remains primarily negative, and oscillates near the static binding energy featuring oscillatory behavior with low amplitude, signifying stable binding. At high temperatures, E_{bind} 's amplitude increases, which is expected because of the increase in thermal energies, but also might indicate desorption.

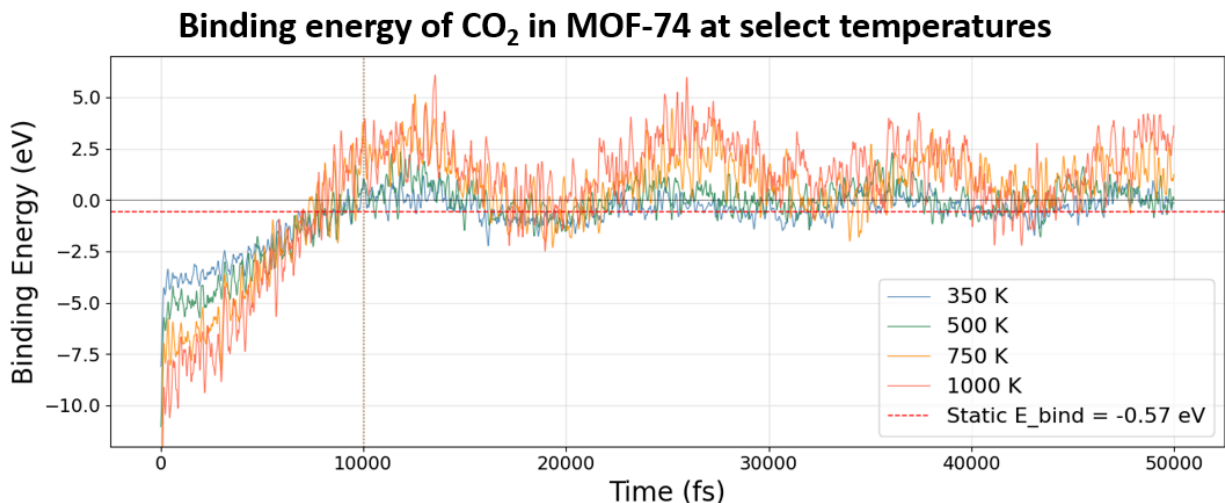


Figure 2: Binding energy of the CO₂ molecule as a function of femto-seconds. As temperature increases, the oscillatory amplitude increases, indicating unstable bonding.

In Figure 3, Mean Binding Energy and Mean Mg-O distance also reflect this same behavior, with MBE, and its standard deviation, increasing with T, indicating reduced adsorption, and reflecting larger thermal motion, while the Mg-O distance increases with temperature, which indicates less binding affinity.

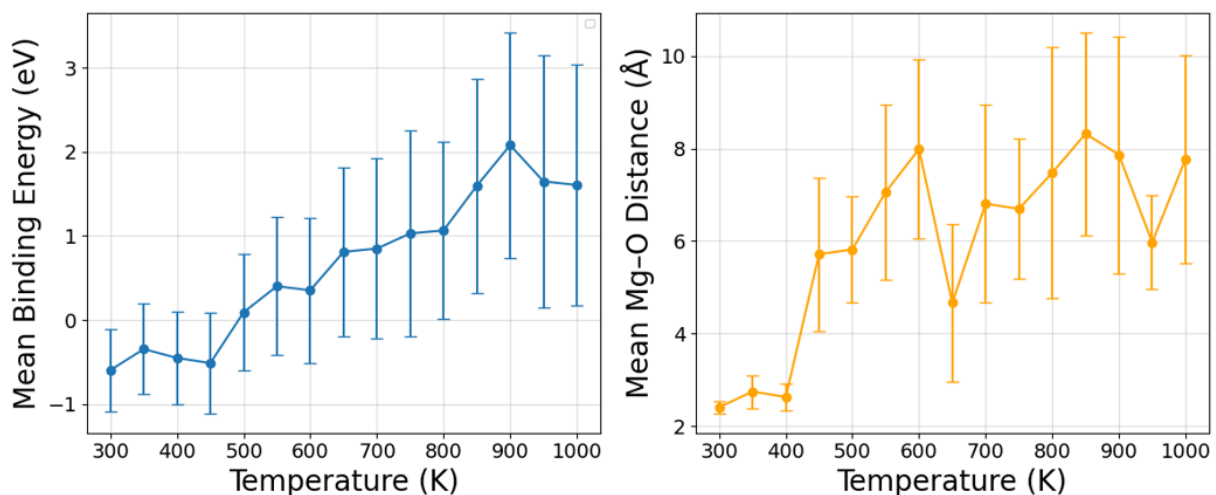


Figure 3: Left plots MBE while the right plots Mean Mg-O distance across the entire temperature sweep.

However, graphs only tell part of the story, and we wanted to know the temporal aspects of the binding. How close should the bond distance be to be considered dwelling? To do that, we set a binding distance of a Mg-O bond as a threshold for measuring total bound time across the temperature sweep, and found that binding time decreases drastically past 450K, indicating migration away from the initial Mg site past the initial Mg site.

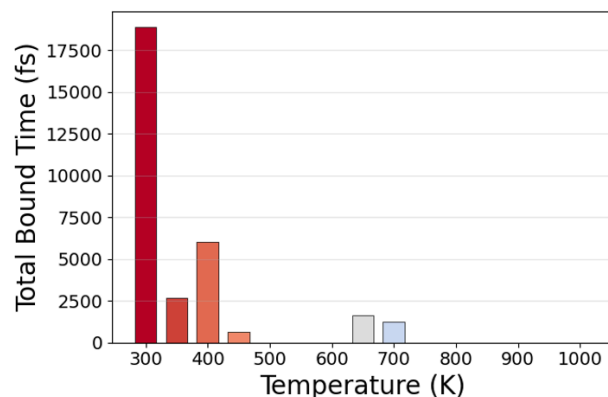


Figure 4: Total Bound Time (fs) plot against the entire temperature sweep.

There is a slight fluctuation at around 650-700K, which probably suggests transient binding likely due to increased thermal motion of both the CO₂ and the MOF framework, that might introduce dipole-dipole and or van der Waal interactions.

With the help of Julia Baratta, we also were able to visualize our trajectories from the different MD trajectories to confirm that as temperature increases, there leads to rapid desorption and adsorption CO₂ rates (hopping behavior), and distance traveled across Mg sites.

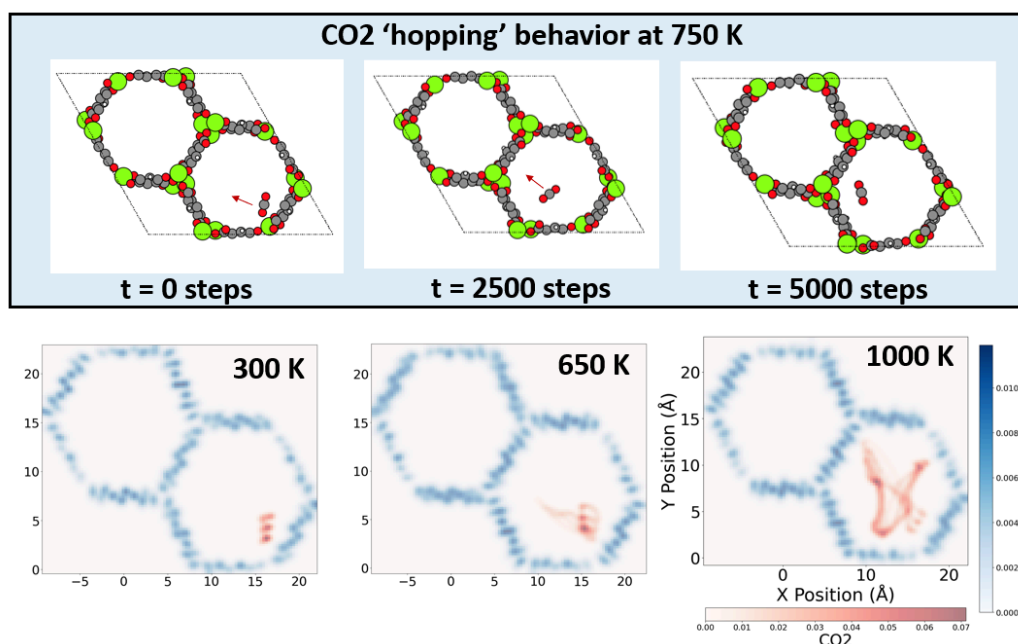


Figure 5: Top shows CO₂ hopping behavior at 750K, showing the transient and unstable binding as temperatures increase. Bottom shows the heat map trajectories, with at low temperatures showing stable and minimal movement while at high temperatures, the hopping behavior increases and the CO₂ molecule moves more.

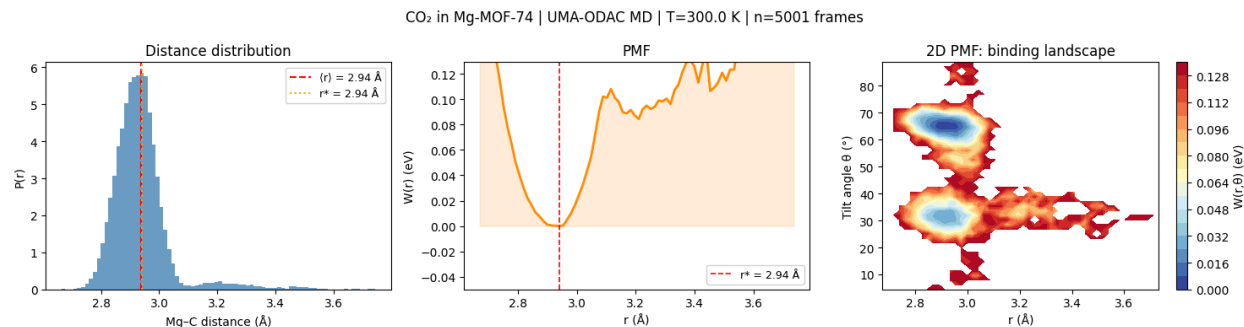


Figure 6: Three graphs were extracted from one temperature step (this case 300K). Left: Mg-C distance distribution centered near 2.94 Å. Middle: PMF showing a free-energy minimum at the preferred binding distance. Right: 2D PMF landscape of Mg-C distance and CO₂ tilt angle, showing preferred adsorption orientations near the Mg binding sites.

Further analysis: Inverted Boltzmann Distribution

To characterize how CO₂ binds within Mg-MOF-74 across temperatures, we applied the inverse Boltzmann relation to MD trajectories, converting the observed probability distribution of Mg-C (measured to the Carbon now) distances directly into a free energy landscape (Potential of mean force: PMF).

From the trajectory files, we extracted three parameters: the most probable binding distance r^* was extracted from the PMF minimum, the mean distance $\langle r \rangle$ was computed alongside it to capture how far CO₂ strays from that site on average, and the difference, $|\langle r \rangle - r^*|$, between these values to represent how “restless” the CO₂ molecule is, and how much probability it actually spends there, due to thermal energy.

Across all temperatures from 300K to 1000K, r^* remained fixed at ~ 2.94 Å, confirming this as the intrinsic Mg-CO₂ binding distance regardless of thermal conditions and where it is at the bottom of the potential energy well. The gap $|\langle r \rangle - r^*|$ grew from just 0.01 Å at 300 K to ~ 0.09 Å near 750 K, reflecting increasing thermal excursions away from the site. The CO₂ tilt angle of adsorption stays consistently around 53-56° across all temperatures, suggesting the preferred orientation is controlled by the MOF's local geometry rather than thermal effects, however future testing with different starting orientations and directions can change this.

Together, these results indicate that Mg-MOF-74 desorbs CO₂ at high temperature primarily through translational delocalization of CO₂ drifting within the pores, rather than any change in the fundamental binding geometry. In addition, this confirms our intuition from the MD section, that at higher temperatures binding does occur, but quickly desorbs and re-adsorbs onto other Mg sites, due to the high global increase in thermal energies that are constantly competing with the MOF's carbon adsorption capabilities, making it harder at higher temperatures for stable CO₂ adsorption capabilities to shine through.

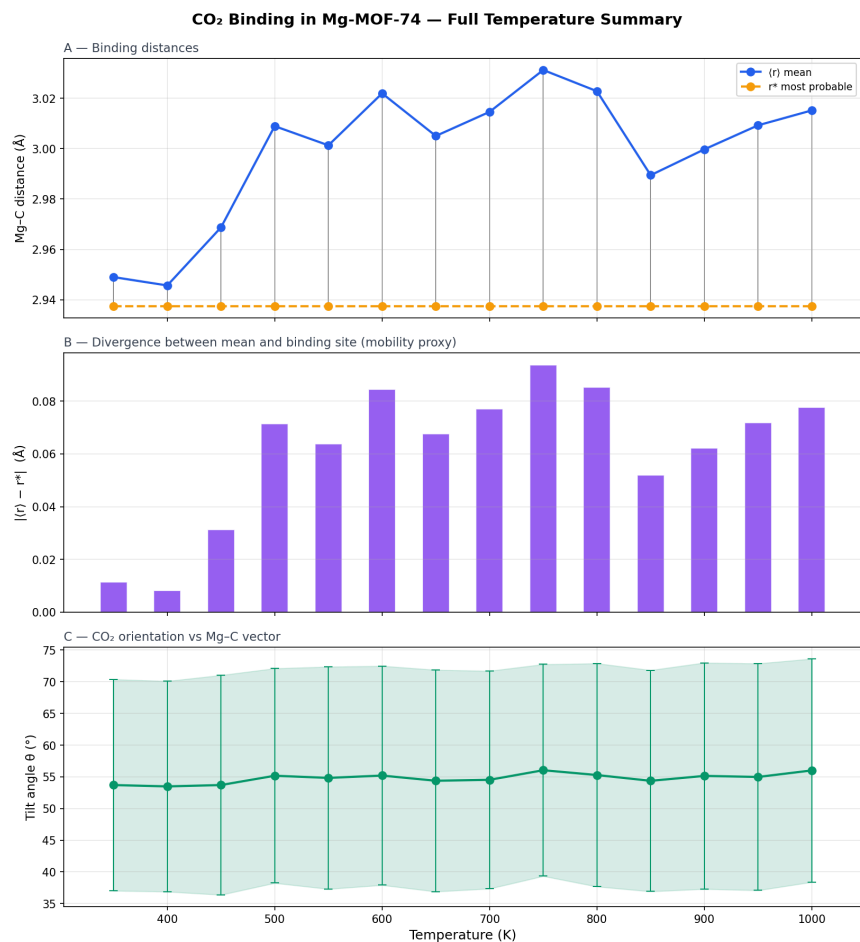


Figure 7: Summary graphs tracking binding distances for $\langle r \rangle$ and r^* (top), $|\langle r \rangle - r^*|$ gap (middle), and CO₂ tilt angle of binding.

Limitations and Future Work

We were limited in our computational abilities, ironically. We thought by using an MLIP, we could cut run time down, and expend less compute power, however, what we did not know is that we used CPU to process our calculations instead of the GPU, which does not adhere to the model, as it does not have a feature to support Linux GPU functionality. By this point, we were already halfway through, and decided to just continue instead of switching over to adroit or another computing cluster (sum cost fallacy). Thus, we are also limited in the iterations conducted. Future work would definitely include more simulations on this same process, in order to get better error bars and see where in the data there are fluctuations, or are they just stochastic anomalies. Furthermore, we would also want to use this mode of analysis to observe thermal dynamics of other MOFs, including CAF-20, which is a more porous MOF specifically designed for carbon capture, and also include the same MOFs but with different organic linkers such as nitros, amines, and other functional groups that enhances affect binding time and binding distance in literatures and experiments [12], but seeing if they can also improve binding abilities in the case of rising thermal temperatures.

Other future work in our analysis, could include a free-energy component. While the inverse Boltzmann PMF captures the preferred binding geometry and its temperature dependence, future work should incorporate enhanced sampling methods such as umbrella sampling to improve statistics at low temperature and obtain absolute binding free energies [13]. Extending the analysis to a full 3D free energy surface and multiple CO₂ loadings would provide a more complete picture of adsorption thermodynamics relevant to real operating conditions.

Compared to experimental methods, desorption was reported to occur in real experiments to be around 353K-380K, however with the help of helium purges to help desorption and at low pressures. This seems to be around the ballpark of what was observed in our simulations (400-450K), keeping in mind the simulation is run with only one CO₂ molecule and with no additional compounds [14]. In future work, it would be interesting to see the same process be done with a compound of molecules, also adjusting pressures, modeling situations such as in direct air capture or even in water capture, to give a more realistic model for application and experimental purposes.

Conclusion

We were able to create a workflow that models the thermal dynamics of CO₂ binding in the Mg-MOF-74 using the UMA-ODAC calculator. Our results from the extracted parameters of binding energy, Mg-O distances, temporal binding threshold, and the inverse Boltzmann analysis yielded that as temperature increases, the CO₂ molecule binding becomes more transient, hopping from metal site to metal site, and the probability of staying at the initial metal location decreases.

Throughout the project, I learned something as simple as running molecular dynamics simulations with MLIPs are still quite time consuming. Of course, we realized too late about the compute limitations (CPU v. GPU), but I still wonder how much time we would have been able to shave off (maybe one day v. 50 hours?), which goes to show how modeling the interactions of atoms and molecules is not a trivial task. I was not expecting the amount of theory and corrections/approximations that are required of MD and DFT coming into the course, and this project provided a good high level overview of the different techniques and computational tricks that modern material scientists can use to downselect experimental labor. I also learned how to incorporate my classmates' techniques and constructive criticism into how I planned my next steps and analyses. Finally, this was one of my first times doing a complete, self derived computational research project, and being able to go through this process was rewarding in itself such that I was able to put to action what was learned during the lectures and homeworks.

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